

Evidence-first selective inference for Bengali hallucination detection

A study of routing, abstention, provenance, and domain shift

Research question

When should exact evidence decide, and when should the system abstain to a language-model judge?

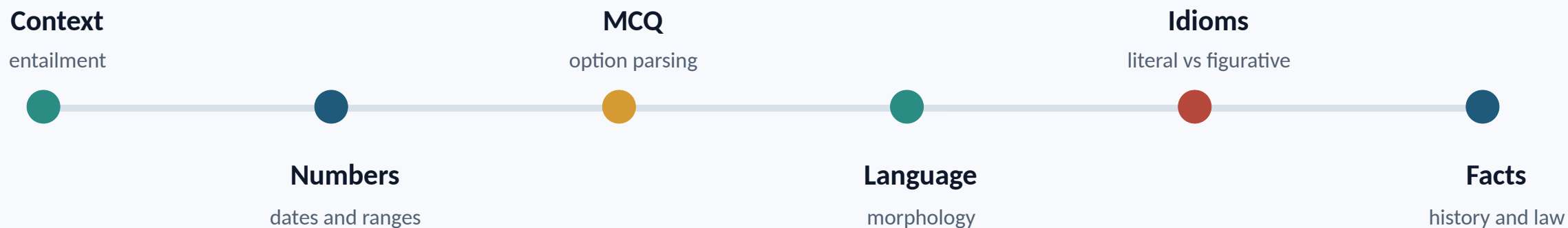
We trace the hypotheses, failed experiments, routing policies, regression gates, and Phase 2 shift.

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One binary label hid many different reasoning tasks

The same label had to cover context faithfulness, language knowledge, and outside facts.



A single judge prompt treated all six as the same problem.

That worked as a baseline, but it was prompt-sensitive and weak on culturally specific relations.

The working hypothesis became a selective cascade

Every stage could answer, or abstain. Abstention was a normal outcome, not a failure.

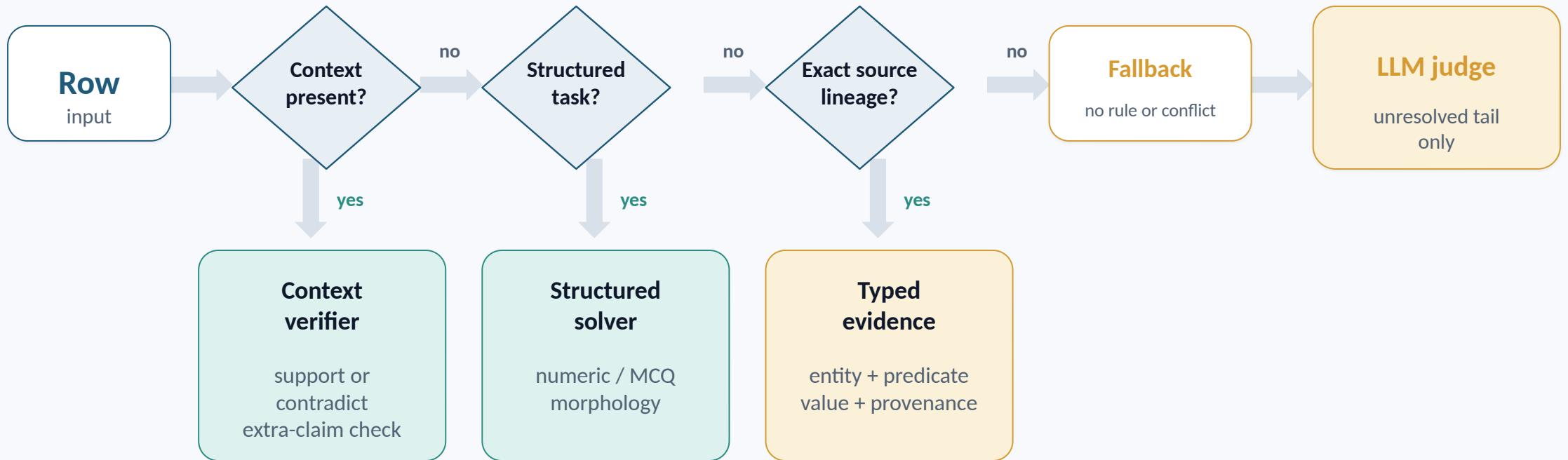


Deterministic stages record the source family and relation behind each decision.

If those fields are not defensible, the row moves forward unchanged.

Routing made abstention an explicit state

A branch could return 0, 1, or abstain. Only abstentions moved to the next branch.

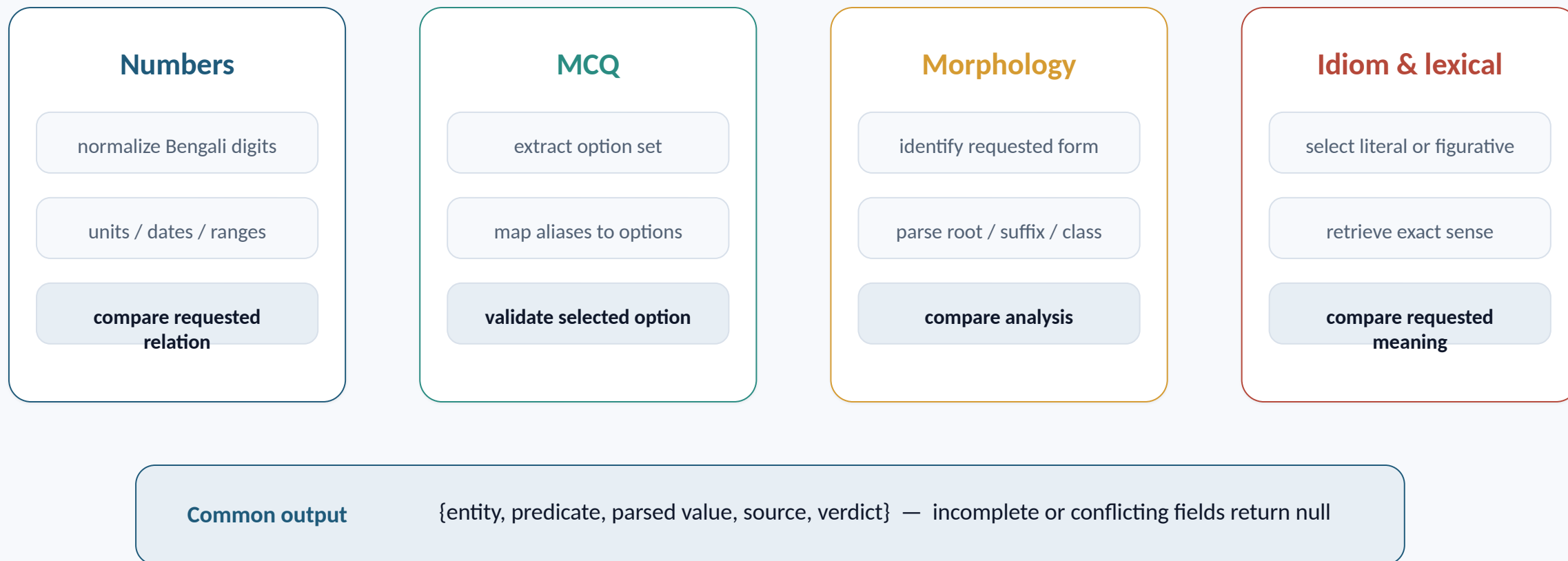


Decision contract

branch(row) -> {0, 1, null}; null means abstain, never a guessed label

Structured solvers shared one decision contract

Each solver parsed a task-specific representation, verified the requested relation, then answered or abstained.



Evidence could decide only when the relation matched

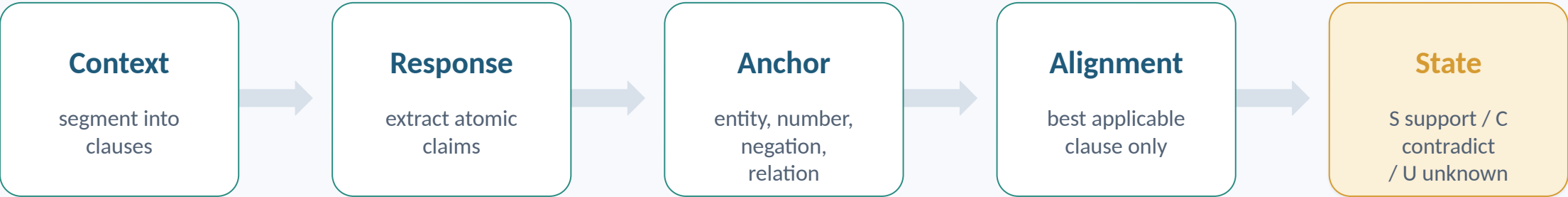
Similarity proposed candidates. It never overruled stronger evidence.

1	Context contradiction	same clause, prompt anchors, no competing claim
2	Exact source lineage	prompt, response, and applicable context agree
3	Structured solver	parsed value, entity, and predicate align
4	Typed evidence	exact idiom sense, law section, or two-source fact
5	Fuzzy retrieval	candidate only; otherwise abstain

Conflict between equal-strength sources also produced abstention.

Context rows were reduced to claim-level entailment tests

The verifier compared atomic response claims against the relevant context clause, not against global word overlap.



Decision rule over response claims $c_1 \dots c_n$

Any contradiction	exists i : $\text{state}(c_i) = C$	label 0
Complete support	for all i : $\text{state}(c_i) = S$; no extra factual claim	label 1
Incomplete alignment	otherwise	abstain

Qwen judged 158 rows, not the whole test set

The fallback was short, deterministic, and biased toward catching hallucinations.

Context row

1 entailment view

Compare the response against the supplied passage.

Any contradiction returns 0.

No-context row

Up to 3 short views

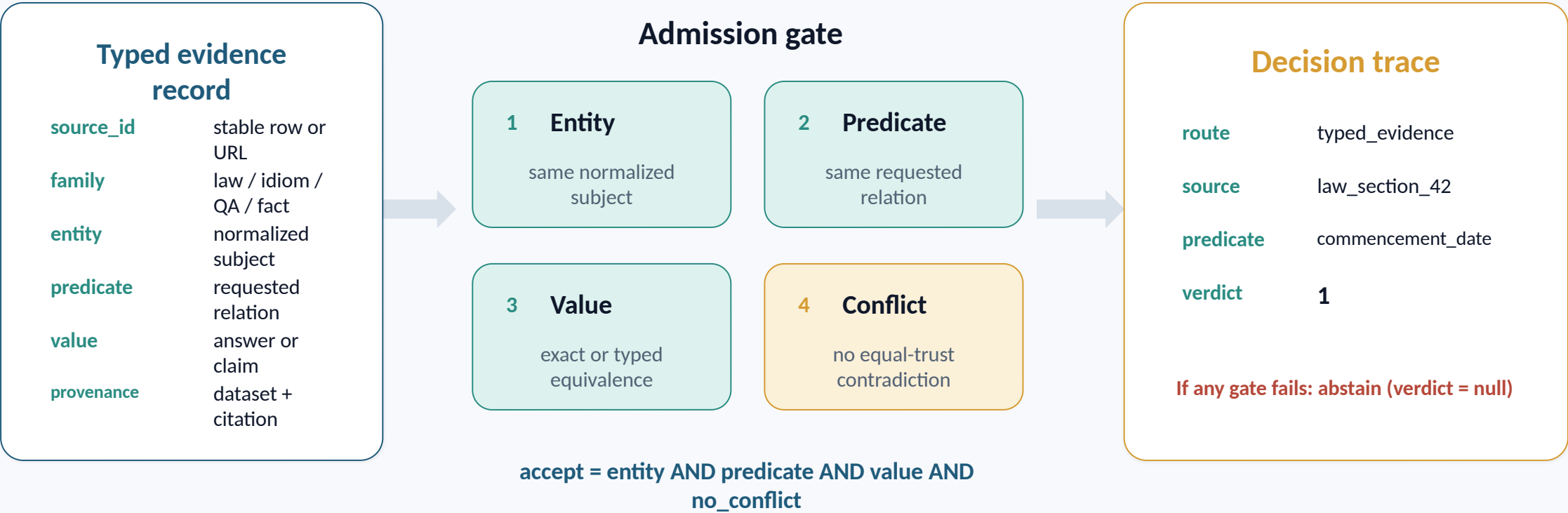
Fact check | exam grading | zero-bias check

Label 1 requires unanimous positive votes.

Qwen3.6-27B Q4_K_P | one-token grammar | fixed seeds and batches

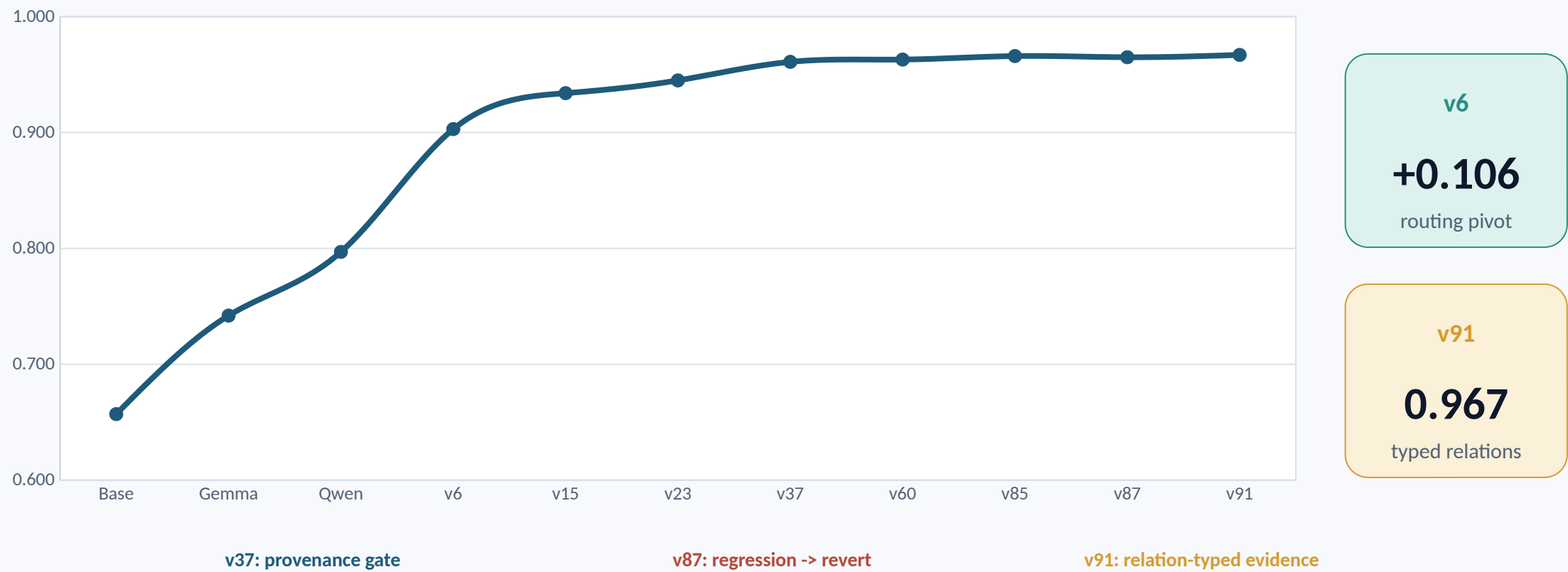
A source record was accepted only after four fields aligned

The trace records why a row was resolved, its source, and the relation that was tested.



Score progression tracked changes in method

Pivots, regressions, and reversions mattered more than a monotonic climb.



Decision logs turned failed runs into routing rules

Each negative result changed what later versions were allowed to decide.

Experiment	Observed result	Decision recorded
Prompt voting	0.851 local, 0.783 public	Stop prompt churn; require new evidence
TF-IDF verifier	0.826 out of fold	Keep as a portable fallback, not a scorer
Raw QA retrieval	Relevant passage, wrong relation	Retrieval proposes; it never decides
v87 fact edits	0.966 to 0.965 public	Require source + predicate; revert

Admission rule: a change must add an evidence trace and preserve the regression suite.

[7] Rejected and scored experiment records in EXPERIMENT.log

Regression gates separated progress from score noise

We kept local correctness, reproducibility, aggregate evaluation, and runtime as separate checks.

249	3	225	0
automated regression tests	byte-identical inference runs	deterministic controls	schema or ID-order failures

Released labels	0.9933 macro F1 on 299 rows
Phase 1 aggregate	0.967 public; 0.982 private
Frozen output	SHA-256 matched the scored CSV
Phase 2 contract	5,000 ordered rows validated in 3:52:02

[7] Regression suite, freeze manifest, evaluation ledger [8] Phase 2 runtime evidence

Phase 2 completed in 3 h 52 min with 95.9% judged

Measured organizer-run inference on 5,000 rows using Kaggle Tesla T4 x2.

5,000

held-out rows

207

deterministic (4.14%)

4,793

Qwen judge (95.86%)

3:52:02

validated end to end

Measured milestones	Elapsed	Observation
Router complete	1:13.18	207 resolved; 4,793 abstained
Model ready	2:51.25	16.332 GiB Qwen across two T4 GPUs
Judge complete	3:52:01.91	13,747.50 s; 0.349 rows/s

42.97% of the 9-hour budget used

5:07:58 headroom

Phase 2 source shift collapsed deterministic coverage

Exact Phase 1 lineage will cover less. The fallback must remain safe when coverage drops.

Web and news

Common Crawl, recent newspapers

Knowledge

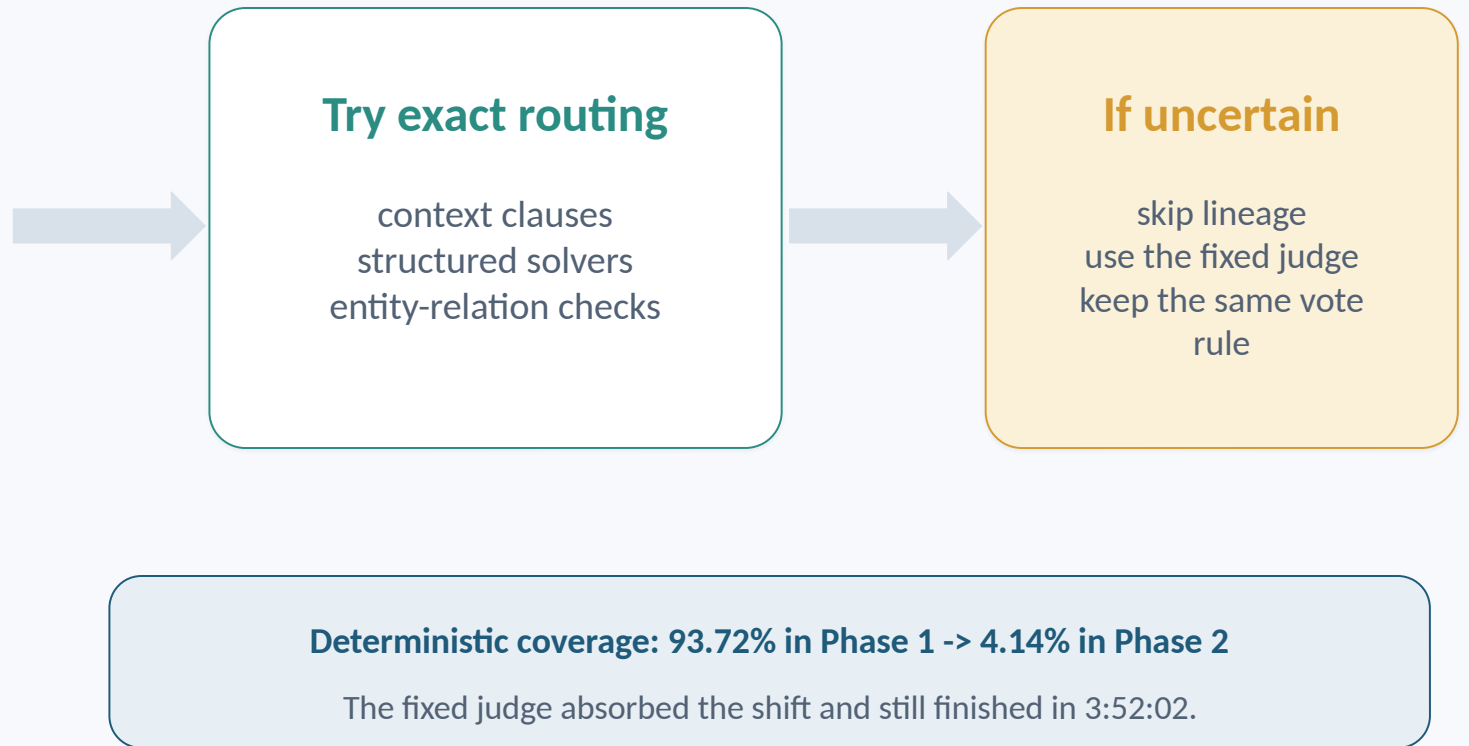
Wikipedia and Banglapedia

Civic and legal

government services and bdlaws

Education and culture

NCTB textbooks and literature



What the experiments support— and what remains open

Supported

Task families need separate policies.
Predicate alignment prevents relation
swaps.
Abstention protected runtime under
shift.

Limitations

Only 299 released labels.
Aggregate splits provide no row-level
errors.
Phase 2 accuracy is still unknown.

Next analyses

Inspect Phase 2 errors when released.
Calibrate the judged tail.
Expand licensed civic and news
evidence.

Questions

Selective inference remained reproducible and completed the held-out run within the runtime limit.